

Summary

Full-stack engineer specializing in building and deploying end-to-end AI/ML systems. Experienced in creating RAG pipelines, fine-tuning LLMs, and architecting scalable backend services in Python and TypeScript with a focus on DX and API design.

Professional Experience

Independent Consultant

Remote

ML Engineer & Product Designer

- Linkby** (Jun 2024 – Nov 2024): Built a unified ML pipeline automating feature engineering and model training from SQL for ads data data from user behavior; enabled parallel RF, SVM, Naive Bayes, LR training with configurable A/B testing.
- Grapple Research** (Nov 2023 – Mar 2024): Improved AI text detection accuracy from 55% to 71% by fine-tuning GPT-3.5 on expanded, deduplicated dataset. Reduced training costs by 64% with truncation & summarizing based on article structure.
- ImmenseX** (Jun 2023 – Aug 2023): Built semantic similarity aggregation tool which reduced the initial prompting time in LLM interactions by 60%. Built production RAG pipeline using Weaviate (BM25/vector search) for conversational AI memory.
- HereAfterLegacy** (Jan 2020 – Feb 2021): Achieved 77% accuracy on sentiment analysis trained on 10K texts (random forest).
- Edelman** (May 2019 – Jul 2019): Trained neural style transfer models for exhibit, applying film-inspired visual filters to selfies.
- Notebrush** (Nov 2018 – Jan 2020): Created user research, animated wireframes, and functional demo for MVP.

Tilting Point

New York, NY

Senior Blockchain Engineer

Nov 2021 – Feb 2023

- Developed ERC-20/721 smart contracts, REST APIs for NFT marketplace, and internal airdropping system. Rescued \$1M+ in MATIC through reverse engineering interactions in deployed contracts.
- Architected blockchain state gRPC microservices, supporting 4+ games with 1000+ events/sec (on/off-chain throughput).

eBay

New York, NY

Software Engineer 2 (Full-Stack)

Jun 2020 – Apr 2021

- Built REST APIs in Scala / Spring-Boot rendering new ads async in real-time from item options selected in UI (<1s latency).
- Built E2E ML delivery infrastructure (K8s + Jenkins) enabling teams to deploy models into sandboxed UIs; accelerated deep learning team iteration by 50% allowing ads team to validate changes without waiting for releases from other teams.

Amelia.ai

New York, NY

Cognitive Implementation Engineer

Jun 2017 – Dec 2017

- Implemented PII redaction and external integrations for enterprise virtual assistant serving Fortune 500 clients.

Selected Open Source Works

- tenets** – Heuristics-based NLP tool to aggregate context for automated LLM prompt generation. github.com/jddunn/tenets
- MagicLogger** – Enterprise Node / browser color logger and universal open style standard. github.com/manicinc/magiclogger
- Restless** - Hierarchical attention network model for anti-malware; 93% accuracy, [ClaMP](https://github.com/jddunn/restless) dataset. github.com/jddunn/restless

Education

- Bachelor of Fine Arts in Design and Technology** (May 2017) – Parsons School of Design

Awards

- 5th Place, AT&T National VR and AR Challenge, with THiNKIMMERSIVE team. (<http://npe100.com/> / mobile app) 2017

Skills

Programming Languages	Python, TypeScript, JavaScript, Go, Scala, SQL, Solidity
Frontend	React, Next.js, Redux, Vue, Nuxt, Tailwind, D3.js, three.js
Backend	FastAPI, Django, Node.js (Express, NestJS), gRPC, GraphQL, RabbitMQ
AI / ML	PyTorch, TensorFlow, Keras, LangChain, LangGraph, LlamaIndex, Transformers, Hugging Face, Scikit-learn, spaCy, Pandas, NumPy, OpenCV, CUDA, XGBoost, Stable Diffusion, Claude, OpenAI
Databases / DevOps	PostgreSQL, Redis, Pinecone, MongoDB, Elasticsearch, AWS (Lambda, EC2, EBS, RedShift), Docker, Kubernetes, Jenkins, GitHub Actions
Tools / Platforms / Other	Unity3D, Adobe Creative Suite, Electron, Capacitor, Android Studio, Quasar, Selenium, Puppeteer